Ankit Panigrahi

Design Engineer

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Whitefield, Bangalore - 560048

Work Experience – 3.5 Years
ankitpanigrahi78@gmail.com

PROFILE

Aerospace Design Engineer with **3.5 years'** experience in **design, development & qualification** of **fuel system LRUs** for combat aircraft (**AMCA, LCA Mk2**) and **test rigs**. Hands-on ownership from **Technical Specification, concept design and theoretical stress calculations** to **GA & detail drawings, ICDs, PDR & CDR** and **prototype / qualification units**, with exposure to **FEA, CFD, thermal & modal analysis**. Skilled in **CATIA V5, NX, SolidWorks, GD&T** and compliance to **MIL-STD, ISO, STANAG and CEMILAC** airworthiness requirements. Collaborates across **Engineering, Analysis, Manufacturing and Business Development** teams; **open to opportunities in India and overseas**.

Naukri : https://www.naukri.com/mnjuser/profile

LinkedIn: https://www.linkedin.com/in/panigrahi-ankit-4b520a209/?isSelfProfile=true

EDUCATION

		Pass-Out
B. Tech	GITA Autonomous College, Bhubaneswar	2023
12th	Times Scholars Gurukul, Bhubaneswar	2019
10th	Army Public School, Kanpur	2017

CERTIFICATIONS

- GD&T Fundamentals & MATLAB Training
- Master Certification Course in CAD & CAM
- CATIA Piping, Tubing & Digital Mockups Training – Dassault Systèmes

TECHNICAL SKILLS

Aerospace Fuel System Design, Development & Qualification	**Aerospace Fuel System LRU Design & Development**, **Concept & System Design**, Modeling, Drafting & **Engineering Drawings**, **GD&T**, **ICDs**, GA & Detail Drawings, **BOM Management**, **Material Selection**, **Weight Budgeting**, Locking Mechanisms, **Spring, O-Ring & Flow Calculations**, **Theoretical Stress Analysis**, **Fault Hazard Analysis**, Reliability, Analysis Team Support (**FEA, CFD, Thermal & Modal**), **PDR & CDR** Documentation, Technical Proposals, **Compliance Matrix**, **Test Procedures (QTP, ATP)**, Qualification Testing Support, Standards: **MIL-STD-810H, ISO, STANAG, ASME Section VIII Div 1**, Airworthiness: **CEMILAC, ARDC, ACP**, **Root Cause Analysis (RCA)**, Process Sheets, KOP, Piping & Tubing, Manufacturing Knowledge & Cost Reduction, Report Making (Microsoft Office).
Computer-Aided Design (CAD)	**Catia V5, SolidWorks**, **AutoCAD**, Creo, NX Unigraphics
Product Life Cycle Management (PLM)	**Enovia**, Teamcenter [Basic]

WORK EXPERIENCE

FCL (Fluid Controls Limited); Aerospace Design Engineer **Dec 2025 – Present**

Project Experience 2	Design and Development of three Fuel System **LRUs** for the **AMCA** (ADA) programme, taken up by FCL: **Quick Disconnecting Coupling (QDC)** for ground pressurisation of the tank, **Non-Return Valve with Thermal Relief Sub-Assembly** and **Engine Feed Line Coupling**; also **Quick Disconnecting Coupling.** **Responsibilities:**

	• Prepared design of **3 major fuel system LRUs** end-to-end, from **Technical Specification** review through **conceptual calculations**, multiple **design concepts** and final concept selection to **assembly and sub-assembly design**. • Designed the **QDC** for ground-condition tank pressurisation, the **NRV** to prevent reverse flow with a **thermal relief sub-assembly** to release tank over-pressure caused by temperature rise, and the **Engine Feed Line Coupling** for direct fuel transfer from Feeder tank to aircraft engine. • Carried out **material selection**, **weight budgeting**, **locking mechanism** design, **Bill of Materials (BOM)** and assembly details. • Performed **theoretical stress, spring design, O-ring and flow calculations** to validate design integrity; prepared **technical compliance** matrices against specification. • Conducted **Fault Hazard Analysis** and supported **reliability** assessment; prepared **Acceptance Test Procedures (ATP)**. • Part of the analysis team for **FEA, CFD, Thermal and Modal Analysis** of the LRUs. • Created **General Assembly (GA)**, detail drawings and **Interface Control Drawings (ICDs)**; developed 3D models using **SolidWorks**. • Completed **Preliminary Design Review (PDR)** of all 3 LRUs with ICDs and detail drawings; **prototype / qualification test units** released for **Critical Design Review (CDR)**.

CTTC (Ministry of MSME); Junior Project Engineer **Apr 2023 – Nov 2025**

Project Experience 1	**<u>CTTC – TDF Projects</u>** • Completed **3-month intensive training** on aerospace manufacturing workflows, from CAD modeling to testing.
Project Experience 1	**Deputed to ADA (Aeronautical Development Agency) for the Design and Development of LRUs [Line Replaceable Units] & System Design.**
	Retractable Probe, Drogue & Relief Valves (LRUs) for **LCA MK2** & Fuel System Pipeline Routing (Modeling) for **Navy & Airforce Directorates**. **Responsibilities:** • Prepared **LRU-level requirements document** for the Combat Aircraft fuel system, ensuring alignment with **ISO, MIL-STD, Stanag** standards and airworthiness requirements set by **CEMILAC** & **ARDC** (Certifying Agency). • Developed detailed **compliance matrices** covering aerospace quality standards, environmental testing. • Designed and developed **combat aircraft fuel system LRUs** in compliance with **MIL-STD-810H**, supporting **Preliminary Design Reviews (PDRs)**, **Critical Design Reviews (CDRs)**, and indigenous product development initiatives. • Created **Interface Control Documents (ICDs)** and developed using **CATIA** and **Enovia**, ensuring full traceability through the **product lifecycle** also prepared **technical reports, presentations.**

Project Experience1	• Performed **theoretical stress analysis** using hand calculations to validate structural integrity. • Produced **2D schematic drawings** in AutoCAD and comprehensive **assembly procedures** in line with aerospace documentation standards. • Designed and simulated **3D CAD models** for functional testing, supporting validation activities. • Prepared **Master Drawing Index (MDI)**, **Kit of Parts (KOP)**, and process sheets for aerospace assemblies. • Developed **Digital Mock-Ups (DMU)** & **Bills of Materials (BOMs)** for design validation and integration planning. • Worked closely with senior engineers and reporting officers, supporting all phases of LRU design, manufacturing, integration, and qualification.
	Universal Test Rig for Valves (LRU Development) & Testing **Responsibilities:** • Designed 2D/3D CAD models for aerospace test rigs (fuel) per **ASME SECTION VIII Div 1**. • Prepared and executed **ATPs** for component approval. • Coordinated cross-functional teams for testing (high/low temperature, altitude, and endurance), setup, and integration issue resolution.
PERSONAL SKILLS	
Others	1. Problem Solving. 2. Client-facing Communication. 3. Project Handling. 4. Teamwork & Quick learner. 5. Presentation Skills.